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ORIGINAL ARTICLE

Worldwide Phylogenetic and Clinical Risk Association of the *Helicobacter pylori* Outer Membrane Gene *hopQ*

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ABSTRACT

Background: *Helicobacter pylori* (Hp) infection is the primary risk factor for gastric cancer, yet the consequences of the infection vary by bacterial and host factors. The outer membrane protein HopQ plays a key role in Hp adhesion, but its genetic diversity and clinical implications are unclear. HopQ–CEACAM interaction facilitates the translocation of the cytotoxin-associated gene A (CagA).

Materials and Methods: We analyzed *hopQ* sequences from 1038 Hp strains, mainly from the *Helicobacter pylori* Genome Project, integrating phylogenetic, genotypic, and disease association analyses. Logistic regression models assessed the association between *hopQ* genotypes and disease, accounting for *cagA* status and Hp ancestry.

Results: *hopQ* genotypes have a strong population structure. The hpAfrica1 and hpNorthAsia (including hspIndigenousAmerica) populations exhibited distinct *hopQ* patterns, suggesting ancestry-specific pathogenicity. *hopQ* genotypes were classified as *hopQI* and *hopQII*; *hopQI* was further divided into high-risk and low-risk variants. Ancestry-adjusted analyses showed that *hopQI* high-risk was associated (odds ratio = 3.1; $p < 0.0001$) with advanced disease (intestinal metaplasia and gastric cancer combined vs. non-atrophic gastritis), independently of *cagA*. In contrast, *hopQII* and *hopQI* low-risk variants correlated with a reduced risk.

Conclusions: The *hopQ* diversity is a key determinant of Hp pathogenicity. *hopQI* high-risk is a predictor of advanced disease, whereas *hopQII* and *hopQI* low-risk may be protective. Understanding *hopQ* variability could improve risk stratification and the targeted therapies for gastric cancer.

1 | Introduction

Helicobacter pylori (Hp) is a gram-negative bacterium that colonizes the human gastric mucosa and is the primary risk factor for developing gastric cancer (GC) [1, 2]. The association between Hp infection and GC is well-established and follows a stepwise progression of gastric inflammatory lesions. The infection typically begins with non-atrophic gastritis (NAG), an early and often asymptomatic inflammatory response to bacterial colonization. Over time, chronic inflammation can variably

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progress to atrophic gastritis and subsequently to advanced preneoplastic lesions, such as intestinal metaplasia (IM) and dysplasia, ultimately increasing the risk of GC [3, 4].

Despite infecting nearly 45% of the world population [5, 6], only a small proportion of individuals develop severe gastric disease. The pathogenic potential of Hp strains is influenced by the presence of specific virulence factors. Among these, CagA and VacA are the best-characterized proteins, playing critical roles in gastric carcinogenesis [7, 8]. Additionally, outer membrane proteins (OMPs) contribute to bacterial adhesion, immune evasion, and host colonization, yet their specific contributions to disease progression remain incompletely understood [9, 10].

Hp encodes multiple OMP families, including Hop/Hor, Hof, Hom, iron-regulated OMPs, and efflux proteins. One of the main families of Hp OMPs is the Hop family [11]. Well-described members of the Hop family include BabA (also known as HopS), SabA (HopP), OipA (HopH), AlpA/B (HopB and HopC), and HopQ. All members of these families act as surface molecules that bind to various adhesion receptors present in the gastric epithelial cells.

HopQ has emerged as a critical factor in bacterial adhesion by interacting with carcinoembryonic antigen-related cell adhesion molecules (CEACAM) receptors. Structural and functional studies have demonstrated that HopQ binds the dimerization interface of human CEACAM1/5 on gastric epithelial cells, serving as an anchoring point for the type IV secretion system, thereby promoting efficient translocation of the oncoprotein CagA into the host cytosol [12, 13]. Importantly, *hopQ* is classified into two major genotypes: *hopQI* and *hopQII* [14]. Previous studies support that only *hopQI* is strongly linked to virulent Hp strains carrying *cagA* and is associated with an enhanced risk of developing GC. Conversely, *hopQII* is more often found in less virulent strains and associated with mild lesions [15]. This genetic distinction underscores that *hopQI* influences bacterial virulence, contributing to gastric mucosal damage and disease progression [16, 17].

Hp has coevolved with humans, giving rise to distinct genetic populations, hpAfrica1, hpAfrica2, hpEurope, hpEastAsia, hpNorthAsia, hpAsia2, hpNEAfrica, and hpSahul, each with unique geographic distributions and virulence profiles [18–20]. *HopQ* variants also exhibit geographic stratification, with *hopQI* predominantly found in Asia and frequently associated with virulent genotypes, whereas *hopQII* is more prevalent in certain regions, including Iran [16, 21].

Despite growing interest in *hopQ*, its genetic diversity and association with disease remain poorly characterized. Although several studies have examined *hopQ* distribution in specific regions, no large-scale, systematic analysis has comprehensively compared its global variation and clinical association across global Hp strains. Moreover, while *hopQI* has been consistently associated with virulence, it has generally been treated as a uniform marker, leaving unresolved whether functionally and clinically meaningful differences exist within this genotype. It also remains unclear whether *hopQ* variants follow population-specific patterns that influence virulence across different geographic and ancestral backgrounds.

To address these gaps, our study leverages a global dataset of 1038 Hp genomes from 50 countries to perform a comprehensive phylogenetic and clinical risk analysis of *hopQ*. We assessed their associations with IM and GC, using NAG as a reference, and showed that *hopQI* can be further stratified into subtypes with distinct clinical behavior, which we classify as high-risk and low-risk variants. *HopQI* high-risk variants are strongly associated with advanced gastric disease, whereas *hopQII* and *hopQI* low-risk variants are associated with reduced risk. We further show that *hopQ* diversity is structured by genetic ancestry and linked to *cagA* status, and identify recurrent genomic duplications of *hopQ*, including both homogeneous and mixed (*hopQI*/*hopQII*) gene copies within the same genome. We also identify evidence of functional divergence, including variation in regions implicated in CEACAM binding, as well as the presence of genomic duplications. Together, these findings highlight the interplay between bacterial genetics, geographic origin, and ancestral factors in shaping Hp evolution, host–pathogen adaptation, and virulence dynamics.

2 | Materials and Methods

2.1 | Identification of the *hopQ* Gene

For our analysis, 1011 Hp genomes from the *Helicobacter pylori* Genome Project (*Hp*GP) [22] and their annotations were downloaded from the NCBI database (BioProject: PRJNA529500). The *Hp*GP set represents single isolates from individuals with various gastric conditions. To identify the *hopQ* gene, amino acid sequences obtained from the coding sequences were scanned for the Hp_OMP motif (PF01856.20) from the Pfam database using the hmmscan tool in HMMER v.3.3.2 [23]. Based on the results obtained, the *hopQ* group was identified, manually reviewed, and the genomic sequences were extracted.

To enhance the representation of Indigenous populations, 27 additional Hp genomes from Peru (BioProject: PRJNA625094) were also included in our analysis. A custom hidden Markov model (HMM) of the *Hp*GP *hopQ* sequences was constructed and employed to search for *hopQ* in the Peruvian genomes, ensuring consistency in gene identification. All genomes represent *de novo* assemblies sequenced by PacBio technology. The genomic characteristics of the 1038 Hp strains are available in Table S1.

2.2 | Classification of *hopQ* Genotypes

For *hopQ* genotyping, the reference sequences *Hp*GP-26,695 (representing *hopQI*; accession number GCA_034695665.1) and Tx30a (*hopQII*; accession number AY147201.1) were used. Alignments were performed using the command-line tool blastn of BLAST+ v.2.15.0 [24]. To ensure reliable genotype classification and avoid biases from partial alignments, only alignments exceeding 1700 nucleotide coverage between the reference sequence and the sequence intended for genotyping were considered. This threshold corresponds to at least 90% of the full *hopQ* gene length, ensuring that most of the gene was analyzed for accurate genotyping. From the BLAST results, the best hit for each sequence against the two reference strains (26,695 and Tx30a) was selected based on alignment length, percentage identity,

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and the minimal number of mismatches. Sequences with identity percentages to their closest reference sequence equal to or greater than 86% were classified as *hopQI* or *hopQII*. Within both *hopQI* and *hopQII*, three distinct subgroups were identified based on sequence identity thresholds: (a) isolates with identity percentages ≥95%, (b) those with identity percentages ≥90% and <95%, and (c) those with identity percentages ≥86% and <90%. These categories were established to reflect genetic divergence within both *hopQI* and *hopQII*.

2.3 | ANIm Analysis of *hopQ* Sequences

To evaluate the genetic similarity among the *hopQ* sequences, Average Nucleotide Identity based on MUMmer (ANIm) values were calculated using pyani v.0.2.13.1 [25]. The analysis was performed in an all-vs.-all manner, with a fragment length (–fragLen) of 100 bp and a k-mer size (–kmer) of 16. ANIm values provide a quantitative measure of sequence identity, which was used to understand the evolutionary relationships and potential functional differences between *hopQ* variants. Results are presented as a heatmap, highlighting clusters of closely related sequences and their levels of nucleotide identity, annotated with population ancestry.

In addition, the ANIm results were used to evaluate both the similarity and the alignment coverage for each sequence relative to two reference sequences, 26,695 and Tx30a. The percentage identity and coverage data were extracted from the respective ANIm output matrices and processed in R using RStudio and the dplyr library. These values were used to validate the previously established classification of *hopQI* and *hopQII* into subgroups (a, b, c) and to further explore sequence clustering patterns.

2.4 | Phylogenetic Analysis of *hopQ* Sequences

Phylogenetic relationships among the *hopQ* sequences were inferred using a multi-step approach. First, multiple sequence alignment was performed using MAFFT v7.525 [26] to align the extracted sequences. A maximum-likelihood phylogenetic tree was then constructed with FastTree v2.1.11 [27]. The tree was annotated and visualized using iTOL v6.9.1 [28], incorporating metadata such as *cagA* status, Hp population ancestry, and *hopQ* genotypes determined in this study. The rooting was set to display the two *hopQ* genotype clusters identified in this study as distinct branches, reflecting their phylogenetic separation.

2.5 | Statistical Analysis

Distributions of *hopQ* genotypes and subgroups in relation to Hp ancestry, *cagA* status, and diagnosis were evaluated using Fisher's exact or Chi-square tests, as appropriate. For *cagA*, genomes carrying a pseudogene were considered negative to reflect a non-functional *cagA*. In particular, we identified 41 strains as *cagA*-positive, but in fact contained *cagA* pseudogenes and lacked other *cagPAI* genes. These strains were reclassified as *cagA*-negative in our analysis, and this final status is reported in Table S1. Graphical representations were performed using GraphPad Prism 9 (GraphPad Software, San Diego, CA, USA).

Hp populations (hps) and subpopulations (hsps) were previously defined by [22]. To improve statistical estimations, we categorized the 17 hps and hsps into eight broader phylogeographic groups for downstream analysis. The definition of these coarser groups was based on phylogeographic criteria. The eight groups were: hpAfrica1 (including hspAfrica1NorthAmerica and hspAfrica1MiscAmericas), hpAfrica2, SWEurope (including hspSWEuropeLatinAmerica, hspSWEuropeChile, and hpNEAfrica genomes), hpAsia2, hpEastAsia, hspNEurasia (combining hspNEurope and hspEurasia), hpNorthAsia (including hspIndigenousAmerica), and hpSahul. The countries included in each phylogeographic group are depicted in Table S1.

Logistic regression was used to calculate odds ratios (ORs) and corresponding 95% confidence intervals (CIs) for the associations between *hopQ* genotypes and disease (outcome variable). To assess heterogeneity across identity subgroups (i.e., a, b, and c), we ran unadjusted models for the *hopQI* and *hopQII* separately for IM, GC, and both outcomes combined. Preliminary analyses of IM and GC performed separately showed comparable effect sizes and consistent directionality across outcomes; therefore, to improve statistical power and model stability, both conditions were combined for subsequent multivariable analyses. In the adjusted analysis of IM and GC ($n=404$) combined (compared with NAG, $n=629$), *hopQ* was modeled as a multilevel categorical variable reflecting the combination of genotype (*hopQI* high-risk, *hopQI* low-risk, *hopQII*) and gene copy number (single vs. duplicate), resulting in ten categories (these categories were modeled independently and included the following combinations: *hopQI* high-risk, *hopQI* low-risk, *hopQII*, *hopQ*-Hardy, *hopQI* high-risk duplication, *hopQI* low-risk duplication, *hopQII* duplication, *hopQI* high-risk + *hopQI* low-risk duplication, *hopQI* high-risk + *hopQII* duplication, and *hopQI* low-risk + *hopQII* duplication). The models were adjusted for ancestry (reference group: SWEurope, selected as it represents the most prevalent haplogroup in the dataset), *cagA status* (reference: Absence; genomes containing a *cagA* pseudogene were considered negative to reflect a non-functional *cagA*), sex (reference: Female), and age groups. As a secondary analysis, we evaluated the disease association restricted to the SWEurope group that represents the largest ancestral population.

Additionally, a multiplicative interaction model was constructed to assess whether *hopQ* modifies the effect of *cagA* in an ancestry-adjusted analysis of IM and GC combined (compared with NAG). The regression model included the categories *hopQI* high-risk, *hopQI* low-risk, *hopQII*, *hopQI* high-risk duplication, and *hopQI* high-risk + *hopQII* duplication, and all *hopQ* x *cagA* combinations were evaluated.

For all described analyses, a p-value <0.05 was considered statistically significant.

2.6 | Identification of Variable Regions in *hopQ*

The identified hopQ sequences were translated into amino acid sequences using the bacterial genetic code. Protein sequences corresponding to *hopQI* high-risk, *hopQI* low-risk, and *hopQII* groups were aligned separately using MAFFT v7.510 [26]. Consensus sequences of HopQI high-risk, HopQI low-risk, and

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HopQII were generated based on the alignments using Jalview [29]. To further refine the alignments, Gblocks v.0.916 was applied with all the combined and aligned HopQ protein sequences to remove poorly aligned and unconserved regions [30, 31]. The Gblocks-trimmed sequences were then separated into HopQI high-risk, HopQI low-risk, and HopQII. Consensus sequences for the three groups were recalculated from Gblocks-trimmed sequences using Jalview. Next, both untrimmed and Gblocks-trimmed consensus sequences were combined with the HopQ sequence of the Hp strain G27 (UniProtKB accession: B5Z8H1) and aligned together with MAFFT to facilitate the identification of variable regions and specific sites of divergence among the groups.

To estimate the effect of the amino acid change in these variable regions in the Gblocks-trimmed sequences, the Kyte-Doolittle hydrophobic score at each site was assigned. A Wilcoxon test was conducted at each aligned position to assess significant differences among the groups, with multiple testing correction performed using the Benjamini-Hochberg False Discovery Rate (FDR) method. Significant positions were visualized through a Manhattan plot, highlighting regions and amino acid sites that distinguish the different HopQ sequence types.

3 | Results

3.1 | Classification of *hopQ* Genes

We identified a total of 1254 *hopQ* sequences in the 1038 Hp genomes; 242 genomes exhibited duplications or triplications of *hopQ*. Additionally, 28 genomes lacked *hopQ*. Thus, the total prevalence of *hopQ* was 97.3%.

Among the 768 genomes carrying a single copy of *hopQ*, 482 were classified as *hopQI* and 278 as *hopQII*. Based on the sequence identity thresholds applied in this study, the distribution of the *hopQI* subtypes was as follows: 212 *hopQIa*, 164 *hopQIb*, and 106 *hopQIc*. All *hopQI* subtypes were associated with the presence of *cagA*. The distribution of the subtypes of *hopQII* was as follows: 110 *hopQIIa*, 162 *hopQIIb*, and 6 *hopQIIc*. The unadjusted disease association analysis showed that *hopQIa* and *hopQIb* were significantly associated with IM and GC, as well as both outcomes combined (Table S2), while *hopQIc* was not. Accordingly, and considering the similar and relatively strong magnitude of associations for *hopQIa* and *hopQIb*, we grouped them as *hopQI* high-risk and designated *hopQIc* as *hopQI* low-risk. All *hopQII* subtypes were classified as low-risk. In eight strains exclusively representing the *hspIndigenousAmerica* population, the gene could not be classified; these strains belong to the Hardy ecospecies, a deeply divergent evolutionary lineage of Hp that shows nearly fixed genetic differences in about 100 key host-interaction genes [32], which were designated *hopQ*-Hardy.

3.2 | ANIm Analysis of *hopQ* Sequences

An all-vs.-all analysis was performed using a MUMmer-based approach to compute the pairwise nucleotide identity among all *hopQ* sequences. The resulting heatmap (Figure 1) displays the pairwise percentage identity values among 1256 *hopQ* sequences, including the two reference strains. Notably, the sequences classified as *hopQ*-Hardy form a cohesive block (designated as O1) that is clearly separated from the other sequences. The other major clade (O2) bifurcates into two branches: R1 and R2. R1 comprises only two sequences with hpAfrica2, which makes sense considering that hpAfrica2 is documented as the oldest ancestry. Furthermore, R2 further subdivides into clades A and B, and within clade A, subcluster A1 comprises all *hopQII* sequences that form a cohesive block, exhibiting high intracluster nucleotide identity and relatively low similarity to the other clades. Whereas A2 constitutes a small, clearly defined cluster composed of *hopQIc* sequences and, interestingly, all in strains with Africa1 ancestry from both continental Africa and from the Americas. In contrast, within clade B, subcluster B1 exhibits greater heterogeneity, including sequences of *hopQIc*, *hopQIa*, and *hopQII*, with numerous cross-links observed among them. Finally, subcluster B2, the largest group, is predominantly composed of *hopQIa*, but also some *hopQIb* sequences. The two reference sequences are grouped within their corresponding clusters. Figure 1 also illustrates correlations of these clades with *cagA* and with disease, showing a strong correlation of B2 but also A2 with *cagA*$^+$ strains. On the other hand, cluster A1 included mostly *cagA*- strains. Furthermore, the two A subclusters, A1 and A2, included mostly low-risk strains, coming from patients with NAG, whereas cluster B2 comprised the high-risk strains coming from most of the patients with either IM or GC. Finally, cluster B1 was a mix of *hopQIc*, *hopQII*, and even *hopQIa*, which, despite being mostly *cagA*$^+$, came from both low- and high-risk strains.

3.3 | Phylogenetic Analysis of *hopQ* Sequences

We performed a phylogenetic analysis of the 1256 *hopQ* sequences, including the *hopQ* gene duplications plus the two reference sequences. The resulting tree (Figure 2) revealed well-defined clades, highlighting the genetic diversity among the analyzed strains. The strains primarily clustered according to their geographical origin, forming distinct clades for regions such as Latin America (Argentina, Brazil, Colombia, Mexico, and Peru), Asia (China, Japan, and Korea), and others. This phylogenetic structure suggests that the distribution of *hopQ* is strongly influenced by geographical origin and ancestry. Regarding *hopQ* alleles, a clear separation between *hopQI* and *hopQII* was observed, indicating a possible association between *hopQ* genotype and strain adaptation to different human populations or environmental conditions. The inclusion of phenotypic information, such as *cagA* status, further revealed potential correlations between these characteristics and phylogenetic groupings. *CagA*$^+$ strains predominantly carried a *hopQI* gene and tended to cluster together, whereas *cagA*$^-$ strains preferentially carried *hopQII* and formed a different clade, suggesting a relationship between *cagA* status and *hopQ* genetic variability. However, there were also exceptions to this rule.

The BLAST analysis revealed that the *hopQI* high-risk group exhibited an average nucleotide identity of 95.2% (range: 93.0%–100%) and an average coverage of 98.7% (median: 99%). In contrast, the *hopQI* low-risk group presented a lower mean identity

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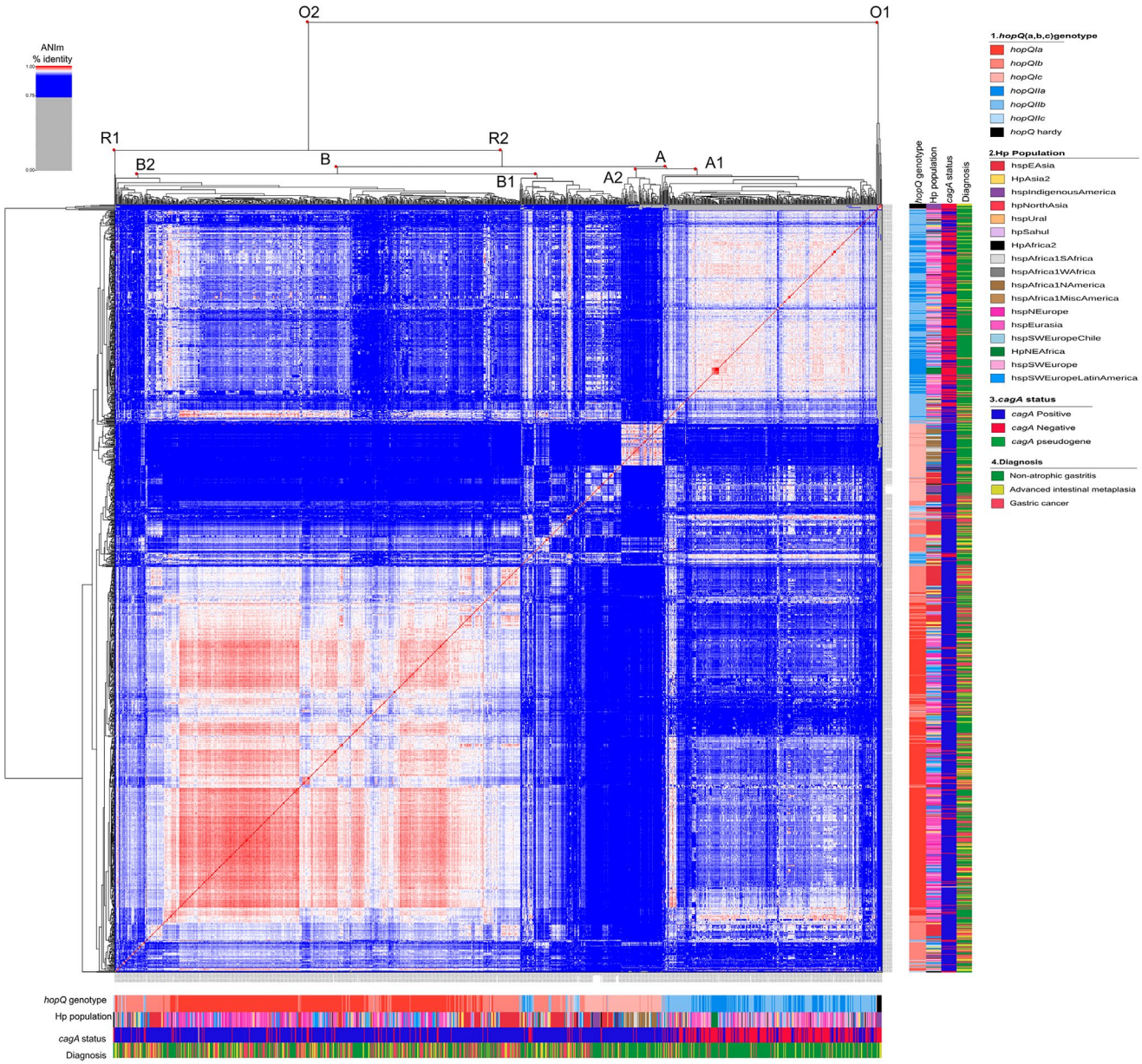


FIGURE 1 | Heatmap of average nucleotide identity based on MUMmer (ANIm) based on *hopQ* alignment. The heatmap displays pairwise percentage identity values among 1256 *hopQ* sequences, including two reference strains (26,695 and Tx30a). Clustering patterns reveal distinct *hopQ* sequence groups, with major clades (O1 and O2) further subdividing into branches R1 and R2. Notably, *hopQ*-Hardy sequences form a distinct cluster (O1) associated exclusively with the hspIndigenousAmerica population. The heatmap highlights nucleotide similarity and genetic relationships among strains.

of 88.4% (range: 86.0%–90.0%) with a median coverage of 96.2%, while the *hopQII* group presented a mean identity of 90.3% and a median coverage of 97.5%. These results are detailed in the Table S3. In the MUMmer-based ANI analysis, the *hopQI* high-risk sequences demonstrated an average nucleotide identity of approximately 95.0% (range: 90.02%–97.81%) and a median coverage of 99%. Meanwhile, the *hopQI* low-risk and *hopQII* groups exhibited average identities of 88.2% and 90.1%, respectively, with corresponding median coverages of 96% and 97%. The BLAST results were corroborated by these MUMmer-based ANI analysis statistics, reinforcing the reliability of our classification scheme (Table S3).

3.4 | Distribution of Genotypes of *hopQ*

Of the 482 genomes with unique sequences identified as *hopQI*, 376 presented *hopQI* high-risk and 106 *hopQI* low-risk, while 278 genomes contained *hopQII*. Out of the 242 genomes that were found to present duplications or triplications of *hopQ*, 138 genomes carried duplications of the same genotype (120 *hopQI* high-risk, seven *hopQI* low-risk, and nine *hopQII*), 104 genomes harbored duplications of different genotypes (92 *hopQI* high-risk/*hopQII* and 12 *hopQI* low-risk/*hopQII*). Finally, two genomes contained triplications of *hopQI* high-risk (Table S4).

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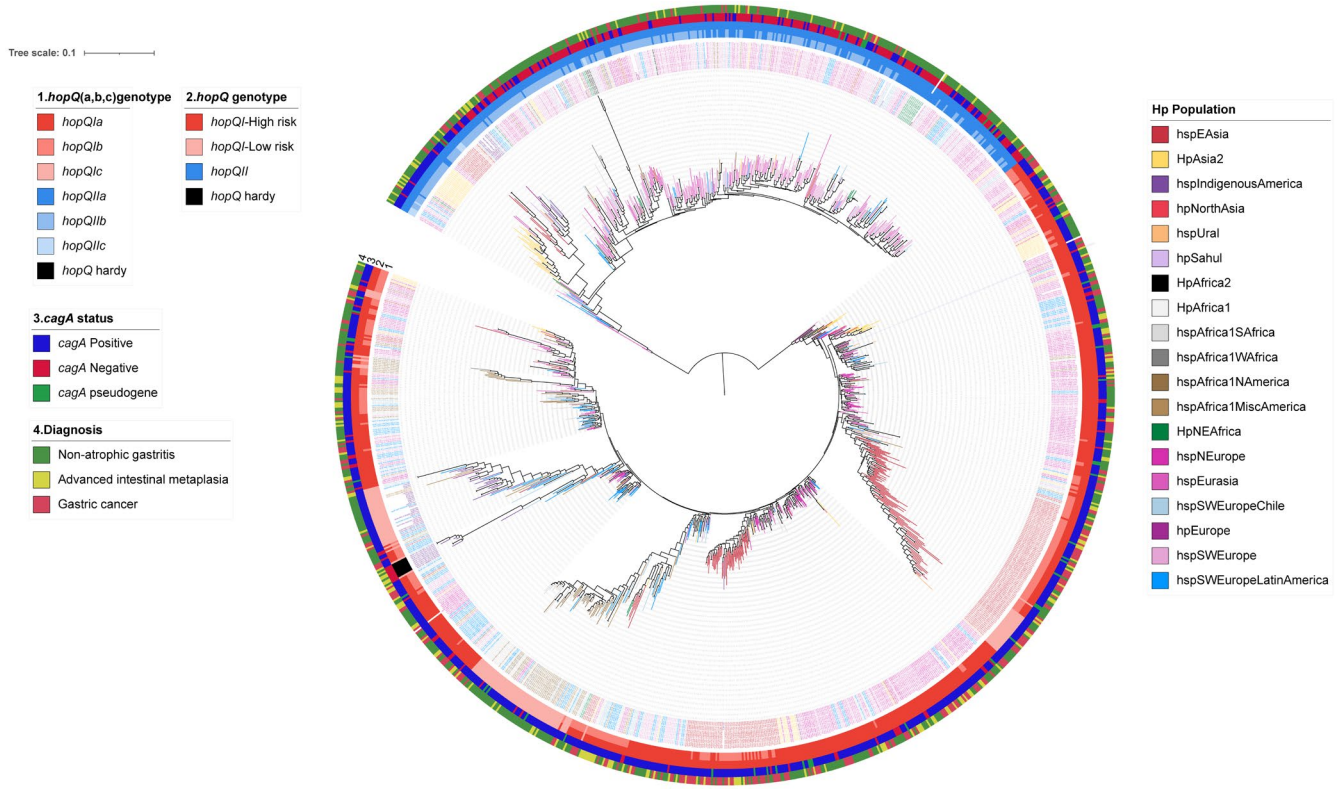


FIGURE 2 | Phylogenetic tree of *hopQ* sequences from Hp, annotated by *cagA* status, *hopQ* types, and population ancestry. The maximum-likelihood tree was constructed from 1256 *hopQ* sequences, including two reference sequences, using FastTree and visualized with iTOL. The phylogenetic structure reveals distinct clades corresponding to geographical origins, with notable clustering patterns for Latin America, Asia, and other regions. Annotations highlight the association between *hopQ* genotype and strain ancestry, where *hopQI* is predominantly found in cagA$^+$ strains, while *hopQII* is more frequent in cagA$^-$ strains. These patterns suggest a link between *hopQ* genetic variability and bacterial adaptation.

Significant differences in the distribution of single-copy *hopQ* genotypes were observed across ancestral populations ($\chi^2 = 451.2$, df = 12, $p < 0.0001$). Populations with a predominance of non-virulent *hopQ* were hpAfrica1 with 51.4% of *hopQI* low-risk and 10.1% of *hopQII*, hpAfrica2 with 100% of *hopQII*, and SWEurope with 60.4% of *hopQII*. Interestingly, hpNorthAsia (including hspIndigenousAmerica) also had the highest frequency of *hopQI* low-risk (47.2%), with *hopQ*-Hardy present in about one-fifth of the strains (Figure 3a). By contrast, hpAsia2 showed a high proportion of *hopQI* high-risk strains (69.2%), and hpEastAsia exhibited an even higher frequency, with 81.3% of strains carrying *hopQI* high-risk. *cagA*$^+$ strains predominantly carried *hopQI* (93.1%), with a strong representation of *hopQI* high-risk (72.2%) and *hopQI* low-risk (20.9%), whereas *cagA*$^-$ strains were primarily *hopQII* (80.9%) ($\chi^2 = 462.4$, df = 3, $p < 0.0001$) (Figure 3b). The distribution of *hopQ* also correlated with clinical outcomes ($p < 0.0001$). In NAG cases, *hopQII* (49.9%) was the most frequent genotype, whereas in IM + GC cases, *hopQI* high-risk was detected in 75.7% of isolates, with only 15% carrying *hopQII*, supporting a strong association between *hopQI* high-risk and advanced disease (Figure 3c). Despite its genetic similarity to *hopQI* high-risk, *hopQI* low-risk did not show the same correlation with IM-GC, reinforcing its classification as a distinct subgroup.

The analysis of *hopQ* duplications also revealed distinct patterns across ancestry ($\chi^2 = 68$, df = 16, $p < 0.0001$). *HopQI* high-risk duplication was the most frequent in the hpAfrica1 (50%), SWEurope (55.6%), and hspNEurasia (57.7%) subpopulations, while the *hopQI* high-risk/*hopQII* combination was predominant in hpAsia2 and hspNEurasia (Figure 3d). No significant associations were found between *hopQ* duplications and *cagA* status ($p = 0.39$, Figure 3e).

Regarding *hopQ* duplications and clinical outcomes, significant differences were observed between NAG and IM + GC cases ($\chi^2 = 14.24$, df = 4, $p = 0.006$), with *hopQI* low-risk/*hopQII* being the most frequent duplication in NAG (46.6%), whereas *hopQI* high-risk duplication was predominant in IM + GC (63.5%) (Figure 3f).

3.5 | Identification of Divergent Regions Between HopQ Groups

Consensus sequences were generated for the HopQI high-risk, HopQI low-risk, and HopQII groups after multiple sequence alignment and trimming with Gblocks. To explore variation, all protein sequences were aligned with the HopQ reference Hp strain G27 (UniProtKB: B5Z8H1), allowing visualization of insertions, deletions, and variable-length segments (Figure S1a). From the Gblocks-trimmed alignment, consensus sequences of similar length were recalculated to better visualize conserved and variable sites (Figure S1b).

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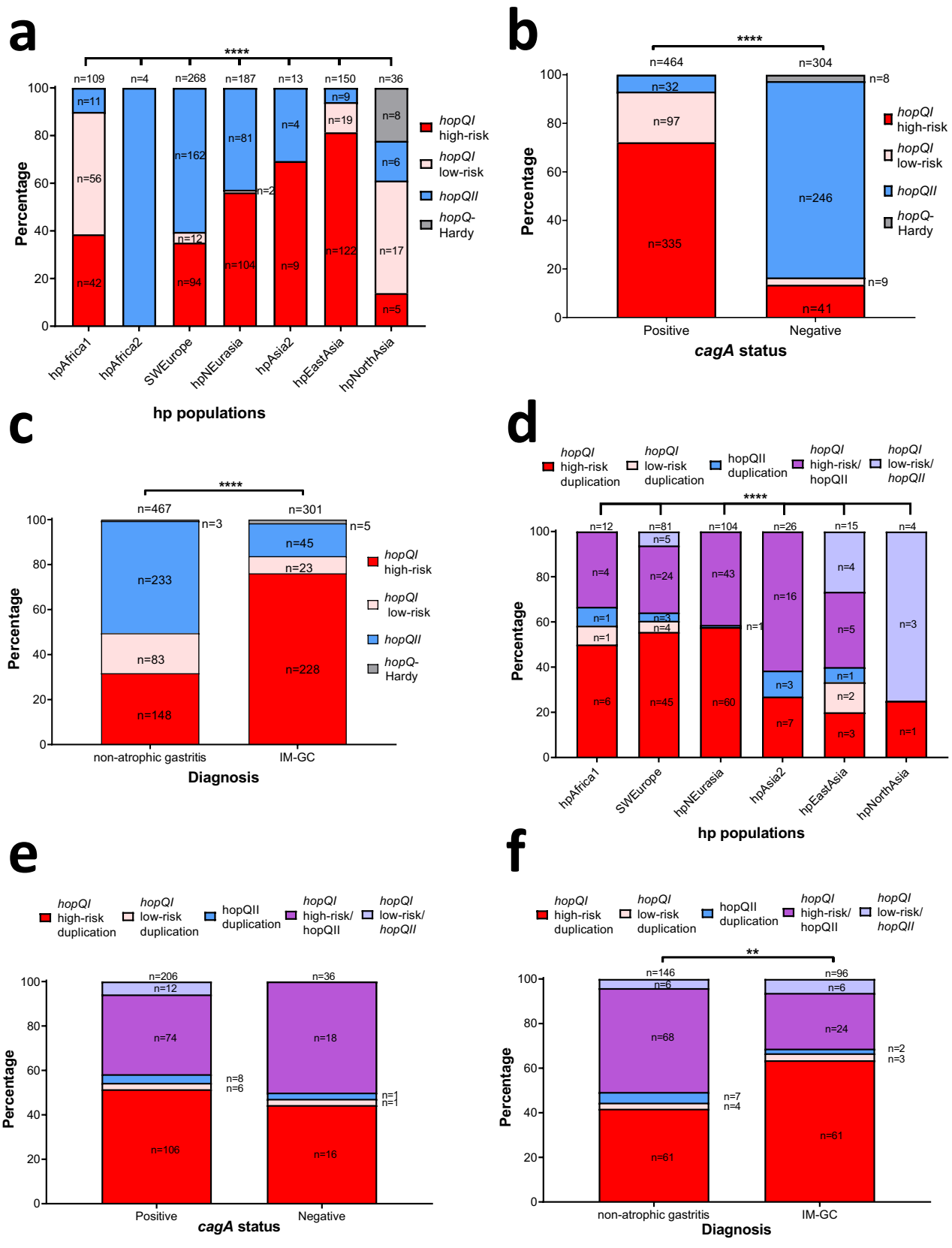


FIGURE 3 | Legend on next page.

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FIGURE 3 | Distribution of *hopQ* genotypes in relation to ancestry, virulence factors, and clinical diagnosis. (a, b, c) Distribution of *hopQ* genotypes among Hp isolates carrying a single copy of the gene, analyzed in relation to (a) ancestry, (b) *cagA* status, and (c) clinical diagnosis categorized as non-atrophic gastritis and intestinal metaplasia and gastric cancer (IM + GC). Distribution of Hp isolates with duplicated *hopQ* in relation to (d) ancestry, (e) *cagA* status, and (f) clinical diagnosis (NAG vs. IM + GC). Bars represent percentages, with the total n for each category indicated above each bar. In (a) and (d), hpNorthAsia (including hspIndigenousAmerica). Significant differences in proportions were assessed using the Chi-square test and Fisher's exact test. Statistical significance is denoted as $^{**}p < 0.001$ and $^{****}p < 0.0001$. No significant differences were observed between *cagA* status and the different types of *hopQ* duplications.

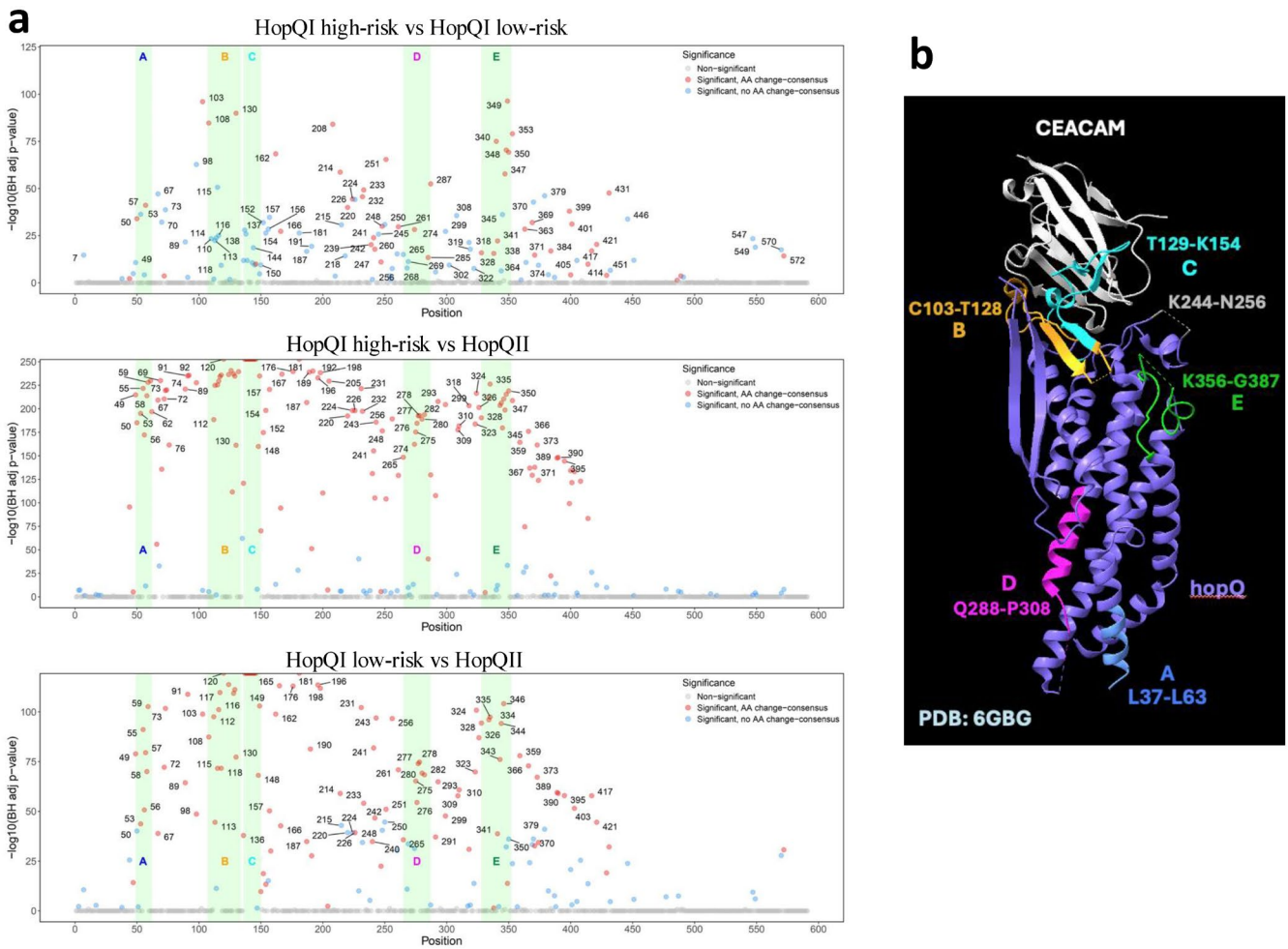


FIGURE 4 | Comparative hydrophobicity analysis of HopQ consensus sequences and structural mapping of variable regions. (a) Manhattan plots showing statistically significant differences in hydrophobicity (Kyte–Doolittle scale) across Gblocks-trimmed aligned positions between HopQI high-risk vs. low-risk, HopQI high-risk vs. HopQII, and HopQI low-risk vs. HopQII. Green shaded areas indicate regions A–E, defined by at least three consecutive sites with significant differences. Red dots represent positions with both statistical significance and amino acid changes in the consensus sequences; blue dots indicate significant positions without amino acid change. Positions in the Manhattan plots correspond to Gblocks-trimmed alignment coordinates. (b) Crystal structure of HopQ in complex with human CEACAM (PDB: 6GBG). Regions A–E are highlighted with the same color code as in panel (a), showing their structural context in loops, helix–loop, or β-strand–loop elements, including sites overlapping the CEACAM-binding interface. Residue numbering in the crystal structure follows the HopQ reference strain G27 (PDB: 6GBG), explaining shifts such as region E (~300–360 in the alignment vs. 356–387 in the structure).

Five regions of high variability (A–E) were identified, each comprising at least three consecutive positions with amino acid differences between groups. These regions are highlighted in green in the Manhattan plots (Figure 4a) and correspond to the same segments mapped onto the HopQ crystal structure (Figure 4b). The positions in the Manhattan plots correspond to Gblocks-trimmed alignment coordinates, whereas in the structural mapping (Figure 4b), residue numbering follows the HopQ reference strain G27 (PDB: 6GBG). This explains apparent shifts, such as region E spanning positions ~300–360 in the alignment but residues 356–387 in the structure.

A Wilcoxon rank-sum test was then applied to the Kyte-Doolittle hydrophobicity values at each aligned position

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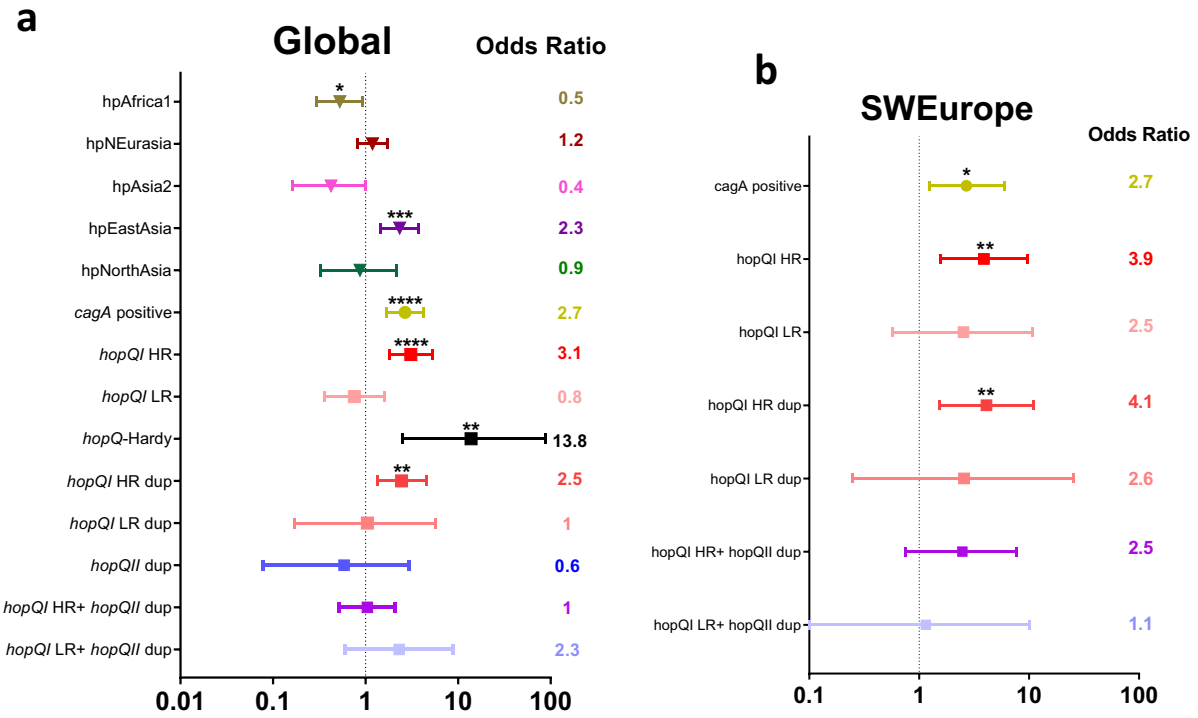


FIGURE 5 | Odds ratio (OR) from logistic regression models assessing the risk of intestinal metaplasia and gastric cancer (IM + GC) compared to non-atrophic gastritis. (a) Logistic regression model including *hopQ* genotype (single-copy and duplications, reference: *HopQII*), *cagA* status (present/absence, reference: Absence), ancestry (hpNorthAsia including hspIndigenousAmerica, reference: SWEurope), sex (reference: Female), and age as predictor variables. (b) Stratified logistic regression model for the SWEurope population, analyzing the association of *hopQ* genotypes (reference: *HopQII*), *cagA* status (reference: Absence), sex (reference: Female), and age with IM-GC risk. The x-axis represents OR values, with the vertical dashed line indicating OR = 1. Error bars correspond to confidence intervals (CI). Statistical significance is denoted as follows: ****$p < 0.0001$, ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$. HR, High risk; LR, Low risk; dup, Duplicate.

using the Gblocks-trimmed sequences. After *p*-value correction using the Benjamini-Hochberg FDR method, 124 statistically significant positions were identified between HopQI high-risk and HopQI low-risk, 187 between HopQI high-risk and HopQII, and 158 between HopQI low-risk and HopQII. Amino acid changes were observed at 53 of 124 positions in the first comparison, 131 of 187 in the second, and 107 of 158 in the third (Figure 4a).

Mapping of the variable regions onto the HopQ–CEACAM structure revealed that Regions B–C overlap with the CEACAM-binding interface, suggesting functional impact. Region E lies adjacent to the binding surface, while A and D, though distal, may influence conformational flexibility or immune recognition (Figure 4b). Together, these findings highlight that HopQ variability concentrates in structurally dynamic, surface-exposed regions, with potential consequences for host adaptation and virulence [33].

3.6 | Multivariate Logistic Regression of Disease Risk

The multivariable analysis confirmed that *hopQI* high-risk was significantly associated with IM + GC (OR = 3.1, 95% CI: 1.8–5.6, $p = 0.0001$), *hopQI* high-risk duplications had a similar magnitude of association (OR = 2.5, 95% CI: 1.3–4.5, $p = 0.004$), and *hopQI* low-risk did not show a significant association (OR = 0.8, 95% CI: 0.4–1.6, $p = 0.46$), and *hopQ*-Hardy also showed a significant association (OR = 13.8, 95% CI: 2.5–88.1, $p < 0.05$), although based on only eight isolates and with a wide confidence interval (Figure 5a). While this finding suggests that *hopQ*-Hardy variants may be of particular biological interest, it should be interpreted with caution due to the limited sample size. As expected, the presence of *cagA* was positively associated with IM + GC (OR = 2.7, 95% CI: 1.7–4.3, $p = 0.0001$).

In the model restricted to SWEurope genomes, *hopQI* high-risk (OR = 3.9, 95% CI: 1.6–9.7, $p = 0.003$) and *hopQI* high-risk duplications (OR = 4.1, 95% CI: 1.5–11, $p = 0.005$) remained significantly associated, whereas *hopQI* low-risk (OR = 2.5, 95% CI: 0.6–10.7, $p = 0.21$) showed no significant association with a wide confidence interval (Figure 5b). The presence of *cagA* was positively associated with IM + GC (OR = 2.7, 95% CI: 1.2–6, $p = 0.013$).

In the interaction model that included ancestry, age, sex, *hopQ* genotype (*hopQI* high-risk, *hopQI* low-risk, *hopQII*, *hopQI* high-risk duplication, and *hopQI* high-risk + *hopQII* duplication), *cagA* status, and their multiplicative terms (*cagA* × *hopQ*), no significant interaction was observed for *cagA* × *hopQ* genotypes (global $p = 0.777$). Overall, these multivariable models support the *hopQI* high-risk allele as a *cagA* independent predictor of IM + GC.

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4 | Discussion

Hp has evolved alongside humans, diversifying into distinct populations with varying virulence potential [22]. Its capacity to colonize the gastric mucosa relies on multiple virulence factors, including HopQ, an outer membrane protein involved in bacterial adhesion to epithelial cells. In addition to mediating adhesion, HopQ contributes to the function of the type IV secretion system, which is required for translocating the oncoprotein CagA into host cells [34].

Using a large and worldwide set of Hp genomes, our study provides new insights into the global distribution, genetic variability, and clinical relevance of *hopQ*. As previously reported, *hopQ* has been classified into two main types, *hopQI* and *hopQII*, with *hopQI* generally linked to greater pathogenic potential, whereas *hopQII* has been considered less pathogenic [14, 35–37]. Our results provide evidence that *hopQ* is a highly diverse gene, and that classification based solely on sequence variants is limited. Our comprehensive worldwide study analyzed both gene sequence type and may be insufficient to capture its clinical relevance. By integrating sequence diversity with disease-associated outcomes, our analysis supports a refined framework in which *hopQ* variants may be more informatively stratified into high-risk and low-risk categories. Importantly, this classification is not randomly distributed but is structured according to phylogenetic clustering and ancestral background.

Our findings suggest that ancestral background influences GC risk, as different ancestries exhibit distinct virulence factor profiles. Thus, the low-risk/A1/*cagA*$^{-}$ cluster (reported as *hopQII*) was formed exclusively by European strains, while the low-risk/A2/*cagA*$^{+}$ cluster (*hopQIc*) included only African strains. Furthermore, the high-risk/B2/*cagA*$^{+}$ cluster (*hopQIa* and *hopQIb*) included Asian and European strains (no African). Finally, B1 was a mixture of high-risk and low-risk, *cagA*$^{+}$ and *cagA*$^{-}$, *hopQIc*, *hopQIa*, and *hopQII* formed by Asian, European, and African strains. These geographic patterns are consistent with a role of human population history and host genetic background in shaping the distribution of *hopQ* variants and their associated risk profiles. Hp has adapted to its host through long-term coevolution of the interacting proteins, particularly adhesin-receptor pairs such as HopQ-CEACAM [38, 39]. The phylogenetic structure observed for *hopQ* mirrors the well-established population structure of Hp, which closely tracks human migration routes out of Africa. In this context, the predominance of low-risk variants in African lineages may reflect a longer history of host-pathogen coadaptation, potentially favoring more balanced host-pathogen interactions despite the presence of canonical virulence factors. In contrast, the diversification of higher-risk variants in non-African populations may be associated with shifts in selective pressures linked to human migrations and population bottlenecks. Additionally, variation in HopQ may be influenced by host genetic diversity, particularly polymorphisms in CEACAM receptors, which are known to differ across human populations and may impose distinct constraints on bacterial adhesion. Together, these observations support a coevolutionary model in which *hopQ* diversity is shaped by the interplay between bacterial phylogeny, human demographic history, and host receptor variability, and in which viral-like adhesins such as HopQ adapt to local CEACAM polymorphisms, helping to explain population-level differences in disease risk.

Notably, our analysis shows that *hopQI* high-risk and *cagA* are independently associated with the presence of advanced gastric lesions. However, this does not preclude the possibility that specific *hopQ* variants may influence *cagA*-dependent mechanisms or more directly contribute to pathogenesis, although this remains to be experimentally validated. The frequent presence of *hopQI* low-risk in strains that also carry *cagA*, yet lack strong disease association, suggests that sequence variation in *hopQ* may be associated with differences in its functional contribution. Although the role of HopQ in facilitating CagA translocation via CEACAM interaction is well established, our results highlight the need for experimental approaches using Hp strains carrying representative *hopQ* variants, including cagA-negative backgrounds, to clarify the extent to which HopQ may independently contribute to gastric pathogenesis.

As a novel finding, we identified *hopQ* gene amplifications in a quarter of Hp genomes. While previous PCR studies detected both *hopQI* and *hopQII* in gastric biopsies [37], they lacked the resolution to distinguish between true gene amplifications within a single Hp strain and coinfections involving multiple strains. Unlike previous studies that relied on PCR-based genotyping, our genomic analysis provided novel evidence of the presence and frequency of *hopQ* duplications and triplications, encompassing either identical or distinct genotypes. These amplification events may confer evolutionary advantages, potentially related to enhanced adaptability or immune evasion. Notably, the observation that *hopQI* high-risk/*hopQII* duplications were not associated with advanced disease may suggest that *hopQII* could influence the virulence associated with *hopQI* high-risk, potentially through mechanisms such as competitive interactions at CEACAM receptor binding or modulation of *cagA* translocation efficiency into the host cells, although no direct evidence is available.

Although our study is based on the largest and most comprehensive genomic collection of *hopQ* sequences analyzed to date, the global diversity of Hp (particularly in the African continent) continues to offer opportunities for further exploration. As our findings are observational, functional validation through protein modeling and in vitro assays will be essential to elucidate the molecular mechanisms and clinical implications underlying the functional differences between *hopQI* high-risk, *hopQI* low-risk, and *hopQII* variants. These insights may contribute to the development of targeted interventions, such as *hopQ*-based diagnostic tools or vaccine candidates. In line with our findings, a recent proteome-wide study identified seropositivity to HopQ as one of the strongest markers associated with the diagnosis of IM [40], reinforcing its relevance not only as a virulence factor but also as a potential biomarker for risk stratification.

An important avenue for future research is the molecular characterization of the differences between *hopQI* high-risk and low-risk variants, particularly in the context of their interactions with *cagA*. Identifying specific sequence variations or structural

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modifications, predictive structural modeling of *hopQ* variants, and functional assays in epithelial cell models should provide deeper insights into how HopQI low-risk may influence CagA activity, ultimately improving diagnostic accuracy and risk stratification for Hp-infected patients.

5 | Conclusion

Our comprehensive analysis suggested that *hopQI* is not a uniform virulence marker but consists of high-risk and low-risk variants with distinct clinical implications. In high-risk populations, such as East Asians, a strong association between *hopQI* high-risk and *cagA* was observed, correlating with increased GC risk. In contrast, in low-risk populations such as Africa, *hopQI* variants may attenuate disease severity. These findings highlight the importance of *hopQ* diversity in risk stratification and provide an outstanding example of evolution, adaptation, ancestry, and virulence in human pathogens.

Author Contributions

Yonne Nasimiyu, Meredith Yeager, and M. Constanza Camargo conceived and designed the study. Edson I. Rubio-Hernández, Difei Wang, and Zilia Y. Muñoz-Ramírez curated and organized the data. Edson I. Rubio-Hernández, Wen Luo, Difei Wang, and Zilia Y. Muñoz-Ramírez performed the formal analyses. M. Constanza Camargo secured funding for the study. Edson I. Rubio-Hernández, Ezequiel M. Fuentes-Pananá, Kaisa Thorell, Roberto C. Torres, M. Constanza Camargo, Javier Torres, and Zilia Y. Muñoz-Ramírez developed and validated the methodology. Ezequiel M. Fuentes-Pananá, M. Constanza Camargo, Javier Torres, and Zilia Y. Muñoz-Ramírez supervised the research process. Edson I. Rubio-Hernández, Ezequiel M. Fuentes-Pananá, and Zilia Y. Muñoz-Ramírez drafted the original manuscript. All authors critically reviewed, edited, and approved the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data on disease status will be made publicly available upon completion of a genome-wide association analysis. However, the information is available from the authors upon reasonable request. All other data supporting the findings of this study are available in the paper and its Supporting Information files.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Multiple sequence alignment of HopQ protein groups and definition of variable regions. (a) Full-length protein alignment of hopQI high-risk, hopQI low-risk, and hopQII sequences, including the reference strain G27 (UniProtKB: B5Z8H1). Divergent blocks are visible across groups. (b) Gblocks-trimmed alignment showing conserved regions of similar length across all groups. Five variable regions (A–E) are indicated, based on clusters of positions with amino acid changes and significant hydrophobicity differences. **Table S1:** Dataset of 1038 *hopQ* sequences analyzed in this study, including genome identifiers, population ancestry, genotype classification, and *cagA* status. **Table S2:** Distribution of *hopQI* and *hopQII* genotypes, including additional subtypes identified within each type, grouped by sequence identity categories: (a) ≥95%, (b) ≥90%–<95%, (c) ≥86%–<90%, with their frequency, *cagA* status, clinical diagnosis, and unadjusted odds ratios for IM, GC, and IM + GC compared with NAG. **Table S3:** Comparative BLAST and ANIm classification of 1254 hopQ sequences analyzed in this study, including genome identifiers, pairwise identity, and coverage against reference sequences (26,695 and Tx30a), and final genotype assignments. **Table S4:** Distribution of *hopQ* subtypes, haplogroups, *cagA* status, and diagnosis in 1038 Hp strains. Percentages for single-copy genotypes (*hopQI* high-risk, *hopQI* low-risk, *hopQII*, and *hopQ*-Hardy), for gene duplications/triplications, and for overall *hopQ* positivity/negativity are calculated within their respective categories; each category sums to 100%.

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